

Supplementary Material: CoralTwin-DT Methods and Tables

Target Journal: Ecological Informatics - Doctoral Research Consortium

Supplementary Material: CoralTwin-DT

Document Title: Supplementary Methods, Tables, and Numerical Formulations for CoralTwin-DT

Target Journal: Ecological Informatics* (Elsevier, Q1)

Manuscript Title: CoralTwin-DT: A Cyber-Physical Digital Twin Coupling Machine Learning and Dynamical Biophysics for Coral Reef Restoration Prioritization under Thermal Stress and Ocean Acidification

Authors: CoralTwin-DT Doctoral Research Consortium

Repository: <https://github.com/HrSly11/CoralTwin-DT.git>

Scientific Attribution: *Resultado obtenido mediante prototipo computacional del gemelo digital.*

Supplementary Methods

Supplementary Method S1: Carbonate Chemistry & Aragonite Saturation (Ω_{arag})

The stoichiometric aragonite saturation state (Ω_{arag}) is calculated using Dickson & Millero (1987) carbonate equilibrium equations:

$$\Omega_{\text{arag}} = \frac{[\text{Ca}^{2+}]_{\text{sw}} \cdot [\text{CO}_3^{2-}]_{\text{sw}}}{K'_{\text{sp}}(\text{arag})}$$

where $[\text{Ca}^{2+}]_{\text{sw}} = 0.01028 \cdot \frac{S}{35.0} \text{ mol/kg}$, and $K'_{\text{sp}}(\text{arag})$ is computed as a function of temperature (T in Kelvin), practical salinity (S), and hydrostatic pressure (P in dbar):

$$\ln K'_{\text{sp}}(\text{arag}) = -171.945 - 0.077993 \cdot T + \frac{2903.293}{T} + 71.595 \cdot \log_{10} T + \left(-0.068393 + \frac{0.0017276}{T} + 88.46 \cdot \frac{S^{0.5}}{T}\right) \cdot S^{0.5}$$

Supplementary Method S2: Optical Water Column Diffuse Attenuation ($K_d(490)$)

Surface downwelling irradiance attenuation at 490 nm is derived from Sentinel-2 MSI Bottom-Of-Atmosphere reflectance (R_{rs}):

$$K_d(490) = 0.0166 + 0.052 \cdot \left(\frac{R_{\text{rs}}(490)}{R_{\text{rs}}(560)}\right)^{-1.24} + 0.008 \cdot \text{Turbidity}_{\text{NTU}}$$

Daily downwelling photosynthetically active radiation (PAR) reaching the benthic substrate at depth z (meters) is modeled via the Lambert-Beer formulation:

$$\text{PAR}(z, t) = \text{PAR}_0(t) \cdot (1 - \rho_{\text{surface}}) \cdot \exp(-K_d(490) \cdot z)$$

Supplementary Method S3: Coupled Dynamical ODE Formulation & Parameter Definitions

The non-linear space-preemption dynamics between live coral cover (C), fleshy macroalgae (M), and algal turf substrate ($T = 1 - C - M$) follow:

$$\frac{dC}{dt} = r \cdot C \cdot (1 - C - M) - d(\text{DHW}, \Omega_{\text{arag}}) \cdot C + \Phi_{\text{restoration}}$$

$$\frac{dM}{dt} = a \cdot M \cdot (1 - C - M) - \frac{g(H_{\text{MPA}})}{M + (1 - C - M)} \cdot M + \gamma \cdot M \cdot C$$

where:

* $r = 0.45 \text{ yr}^{-1}$: Coral intrinsic lateral growth rate.

* $a = 0.6$

Supplementary Tables

Supplementary Table S1: Global Benchmark Reef Monitoring Stations Metadata

Supplementary Table S2: Complete Machine Learning Hyperparameter Grid

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Supplementary Figures & Code

All source code, simulation routines, and high-resolution figures are distributed in the repository under open licenses:

***Master Replication Pipeline:** `run\_all.py`

* **Bioph